

Time Series Modelling Case Study

Gas Price Forecasting Using ARMA and Prophet Models

Git-hub Link: [Time-Series-Modelling-Case-Study](#)

Student Name: Sai Vara Prasad Lekkalapudi

Student_id:23076885

1. Introduction

Forecasting gas prices is important because changes in fuel costs can affect businesses, households, and wider economic activity. Sudden increases or decreases in energy prices can influence operating costs, consumer spending, and financial planning. As a result, organisations often rely on forecasting techniques to better understand possible future price movements and support decision-making.

Time series forecasting uses historical observations to identify patterns that may continue into the future. Different forecasting models are available, each with strengths and limitations depending on the characteristics of the data. After establishing that the series was stationary, the ARMA model was fitted to the gas price data to produce forecasts. More recently, Prophet has gained popularity as a forecasting tool because of its ability to model changing trends while providing measures of uncertainty.

This study examines a dataset containing 500 daily gas price observations collected between January 2025 and June 2026. The analysis investigates the behaviour of the series, assesses whether transformations are required to achieve stationarity, develops an ARMA forecasting model, evaluates Prophet as an alternative approach, and compares the predictive performance of both methods using commonly used forecasting accuracy measures.

2. Dataset Overview and Exploratory Data Analysis

An initial review of the dataset was undertaken before model development to ensure that the data were reliable for forecasting. Missing observations, duplicate records, repeated dates, and inconsistencies in the daily timeline were examined. No major quality issues were identified, indicating that the dataset could be used without substantial preprocessing.

	price
count	500.000000
mean	57.876340
std	16.402565
min	25.440000
25%	41.685000
50%	61.295000
75%	69.347500
max	86.660000

Table 1. Summary Statistics of Gas Prices

Summary statistics showed that gas prices varied considerably throughout the observation period. While the average price was approximately 57.9 units, individual observations ranged from around 25 units to almost 87 units. This level of variation suggests that gas prices experienced meaningful fluctuations over time and therefore present an interesting forecasting problem.

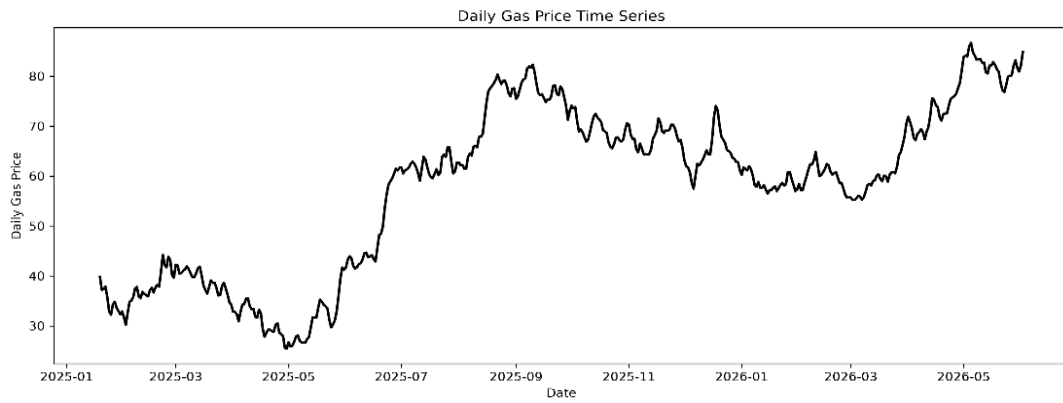


Figure 1. Daily Gas Price Series

Visual inspection of the time series revealed that prices did not remain around a constant level throughout the study period. Instead, periods of growth and decline were observed, with a noticeable increase during the middle portion of the dataset. This behaviour suggests that the underlying characteristics of the series may have changed over time.

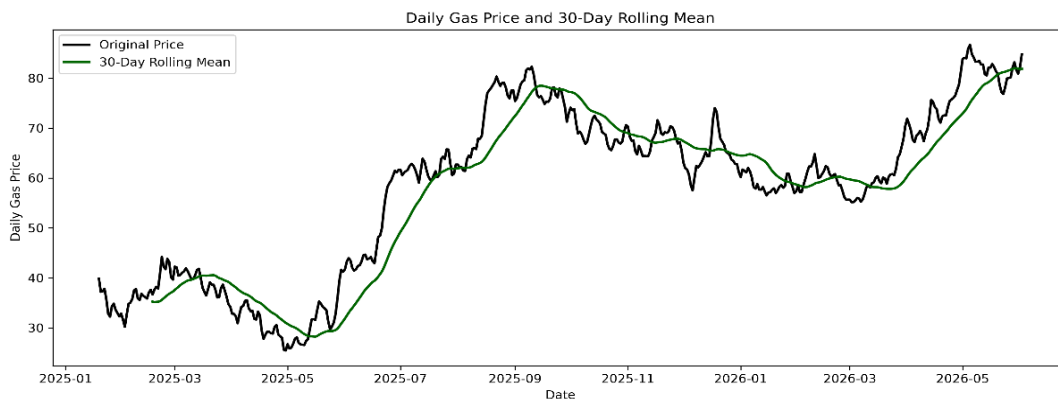


Figure 2. Rolling Mean

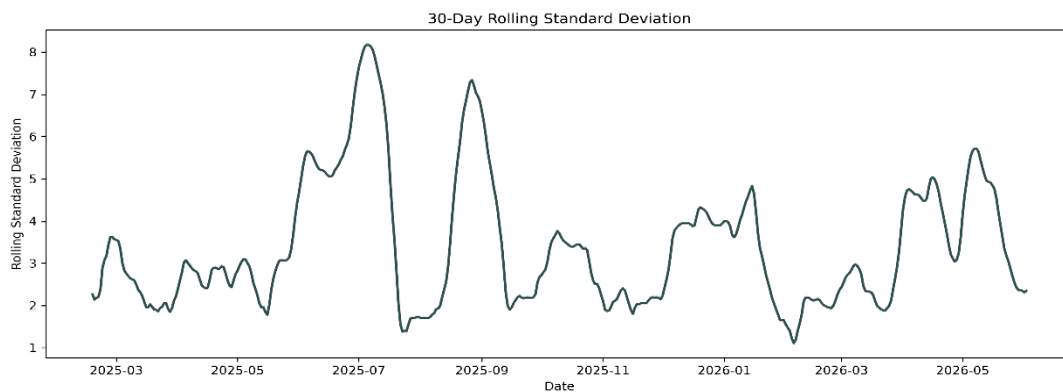


Figure 3. Rolling Standard deviation

To investigate this further, rolling mean and rolling standard deviation statistics were calculated. Both measures showed noticeable variation across the observation period rather than remaining constant. These patterns provided an initial indication that the original gas price series was unlikely to satisfy the stationarity assumption required for ARMA modelling.

3. Assessing Data Stability

To ensure that the data were suitable for ARMA modelling, the stationarity of the original gas price series was evaluated using the Augmented Dickey-Fuller (ADF) test (Dickey & Fuller, 1979).

Series	ADF Statistic	p-value	Conclusion
Original Series	-0.964	0.766	Non-stationary
First-Differenced Series	-7.112	3.91×10^{-10}	Stationary

Table 2. Results of the ADF Stationarity Assessment

The results indicated that the initial dataset was non-stationary, as the p-value was substantially higher than the conventional 0.05 significance level. This finding was consistent with the patterns observed during exploratory analysis, where changes in both trend and variability were visible.

To address this issue, first-order differencing was applied. Rather than analysing the absolute price values, differencing focuses on the change between consecutive observations.

The ADF test was repeated after first-order differencing and produced a p-value of 3.91×10^{-10} . This result suggested that differencing had successfully stabilised the series. As a result, the differenced data were used for ARMA modelling.

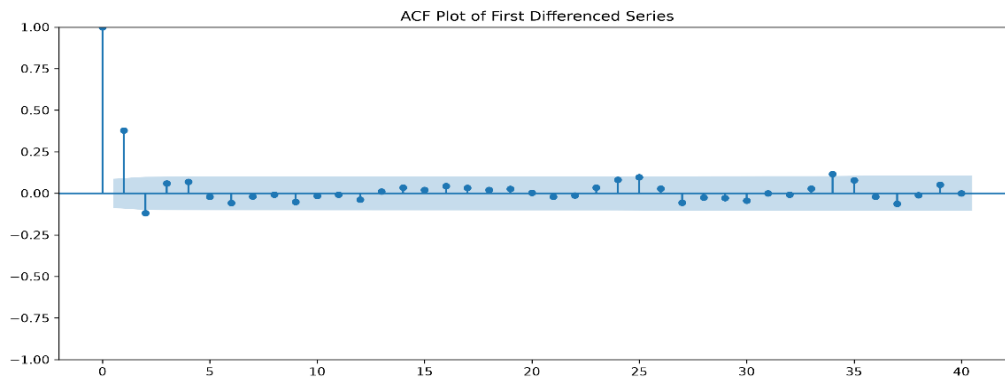


Figure 4. Correlation pattern in Differenced Gas Prices

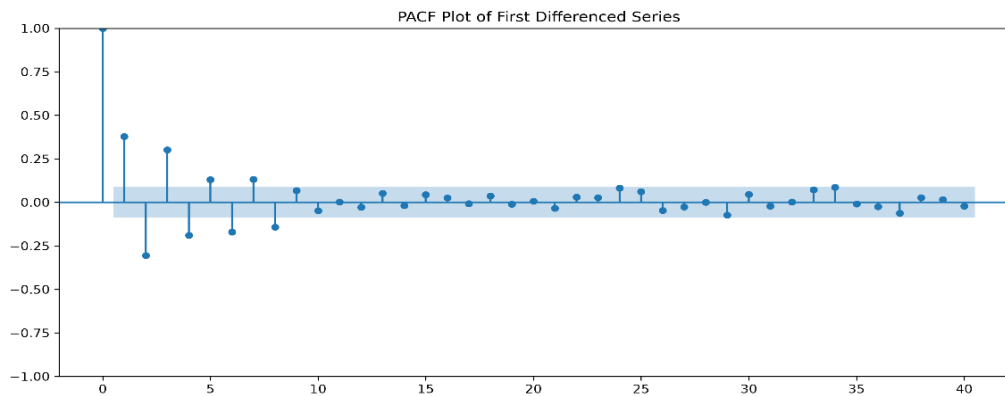


Figure 5. Partial Autocorrelation pattern in Differenced Gas Prices

The ACF and PACF plots of the differenced gas prices were then examined to identify remaining autocorrelation patterns. Several early lags remained significant, suggesting that recent observations continued to contain useful forecasting information. Overall, the results indicated that first-order differencing successfully prepared the series for ARMA modelling.

Taken together, the rolling statistics, ADF test results, and autocorrelation analysis showed that first-order differencing successfully transformed the gas price series into a stationary form suitable for ARMA modelling.

4. Forecasting Methodology

A training and testing approach was used to evaluate forecasting performance. The first 440 observations were allocated to model development, while the final 60 observations were reserved for testing. Keeping part of the dataset separate allowed the models to be evaluated on unseen observations, providing a better indication of how they would perform when forecasting future gas prices.

4.1 ARMA Model Development

After confirming that first-order differencing had produced a stationary series, an ARMA model was developed using the transformed data. ARMA was selected because it is well suited to modelling short-term dependencies within stationary time series and has been widely applied in economic and energy forecasting studies (Box et al., 2015).

To identify the most suitable model, a grid search was carried out across different combinations of p , d and q values. Model selection was based on the Akaike Information Criterion (AIC), where lower values indicate a better balance between model fit and complexity (Akaike, 1974). Once stationarity had been achieved through first-order differencing, the final ARMA(5,8) model was fitted to the transformed series.

	p	d	q	AIC
0	5	1	8	1221.652511
1	2	2	8	1223.250971
2	2	1	7	1223.937871
3	5	1	7	1224.604992
4	3	1	7	1224.916640
5	2	2	7	1225.026932
6	5	2	8	1225.070144
7	3	2	8	1225.132594
8	0	1	8	1225.145420
9	3	1	8	1225.638444

Table 3. Best ARMA Model Based on AIC

The search identified (5,1,8) as the optimal specification, producing the lowest AIC value among the candidate models. Since first-order differencing had already been applied to achieve stationarity, the final forecasting model was implemented as an ARMA(5,8) model on the differenced series.

This result suggests that both recent price movements and historical forecast errors contributed useful information for predicting future gas price changes. After choosing the optimal model, it was trained using the available training data and then applied to forecast gas prices in the test set.

4.2 Prophet Model Development

Prophet was implemented as an alternative forecasting approach to compare its performance with the ARMA model. Unlike ARMA, Prophet does not require stationary data and is designed to capture changes in trend while also providing prediction intervals (Taylor and Letham, 2018).

Several `changepoint_prior_scale` values were explored to determine the most suitable Prophet model. This parameter affects how flexibly the model responds to changes in the underlying trend. Lower values favour smoother trend patterns, while higher values allow the model to adapt more quickly to fluctuations in the data.

	<code>changepoint_prior_scale</code>	MAE	RMSE	MAPE (%)
0	0.50	14.882736	15.698373	18.746497
1	0.01	16.711107	17.724668	21.017973
2	0.10	17.328698	18.182322	21.853586
3	0.05	18.193794	19.049441	22.957280

Table 4. Prophet Hyperparameter Tuning Results

The hyperparameter tuning results identified a `changepoint_prior_scale` value of 0.50 as the optimal configuration. This setting produced the lowest forecasting error across the candidate models and was therefore selected for the final Prophet implementation.

4.3 Forecast Evaluation

MAE, RMSE, and MAPE were used to compare the forecasting accuracy of the ARMA and Prophet models. Lower error values generally indicate better predictive performance. As each metric provides different information about forecasting errors, all three measures were considered during model evaluation (Hyndman & Athanasopoulos, 2021).

5. Results and Model Comparison

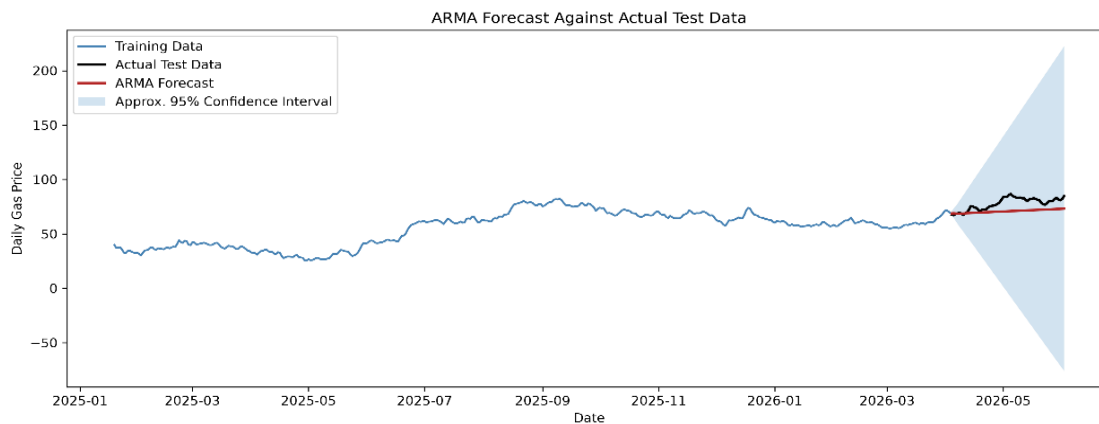


Figure 6. ARMA Forecast Compared with Actual Values

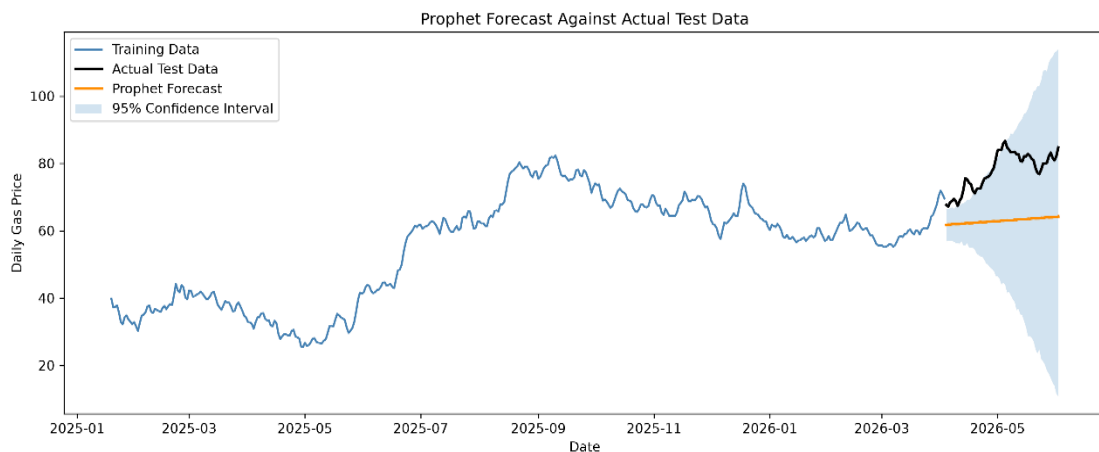


Figure 7. Prophet Forecast Compared with Actual Values

ARMA tracked the observed gas price movements more closely, whereas Prophet produced smoother forecasts that were less responsive to short-term fluctuations. This resulted in better forecasting accuracy for the ARMA model.

	Model	MAE	RMSE	MAPE (%)
0	ARMA on First Differenced Series	7.180442	8.362644	8.880741
1	Prophet	14.882736	15.698373	18.746497

Table 5. Forecast Accuracy Comparison

The forecasting results were consistent with the visual comparison of the models. ARMA achieved lower forecasting errors than Prophet, recording an MAE of 7.18, an RMSE of 8.36, and a MAPE of 8.88%. These results indicate that ARMA provided more accurate predictions during the test period.

The better performance of ARMA is likely related to the way the gas price data behaved after differencing. Short-term patterns were still present in the series, and ARMA was able to capture these changes more effectively than Prophet. As a result, ARMA produced more accurate forecasts during the testing period.

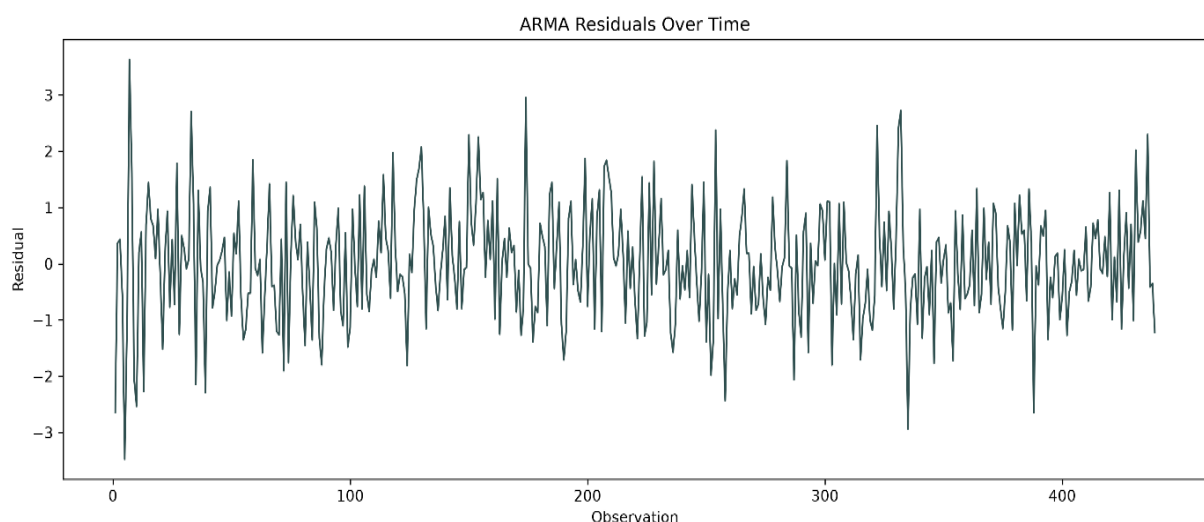
Although Prophet is a flexible forecasting method, its strengths are often more apparent when clear trend changes or strong seasonal patterns exist. In this study, seasonality was relatively weak, which limited the benefits of Prophet's trend-based approach and contributed to larger forecasting errors.

Overall, the results suggest that model performance depends on how well a method matches the characteristics of the data. For this dataset, the ARMA(5,8) model fitted to the first-differenced series produced the most accurate forecasts and was therefore selected as the preferred forecasting model.

6. Residual Diagnostics, Future Forecasting and Limitations

6.1 Residual Diagnostics

Residual diagnostics were carried out for the final ARMA(5,8) model to evaluate whether the important patterns in the gas price data had been adequately captured. This analysis also helped determine whether any systematic structure remained in the residuals.



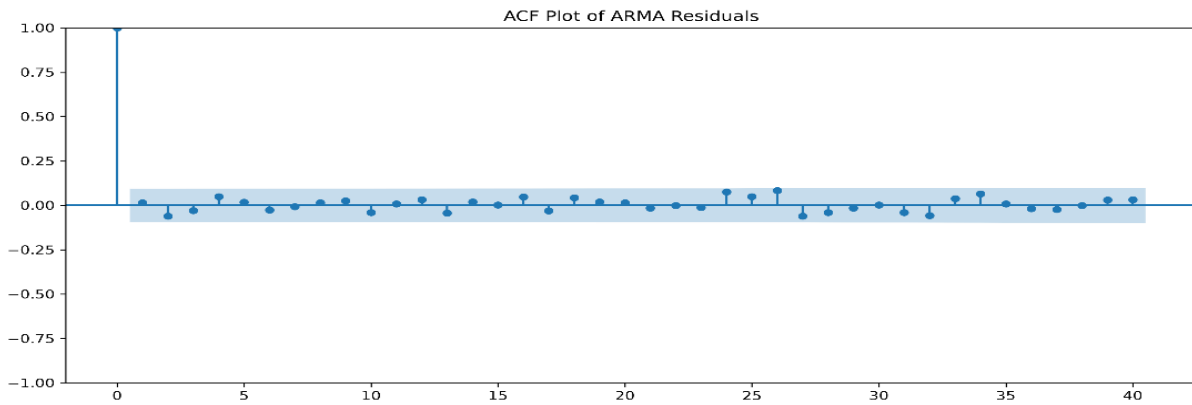


Figure 8. Residual Diagnostics for ARMA(5,8)

The residuals fluctuated around zero with no obvious trend or recurring pattern. This indicates that most of the information contained in the series was captured by the model and that little systematic structure remained unexplained.

Once the model had been fitted, the residuals were examined using the Ljung-Box test to check for any remaining autocorrelation (Ljung & Box, 1978).

	lb_stat	lb_pvalue
10	4.820930	0.902814
20	8.951754	0.983459

Table 6. Ljung-Box Test Results

The Ljung-Box test produced p-values greater than 0.05 at both lag 10 and lag 20, indicating no statistically significant evidence of residual autocorrelation. Together with the residual plots and ACF results, this suggests that the ARMA model adequately captured the main temporal patterns present in the differenced series and was suitable for forecasting.

6.2 Future Forecasting

Following model assessment, the final ARMA model was re-estimated using the complete dataset to generate forecasts covering the next 24 months.

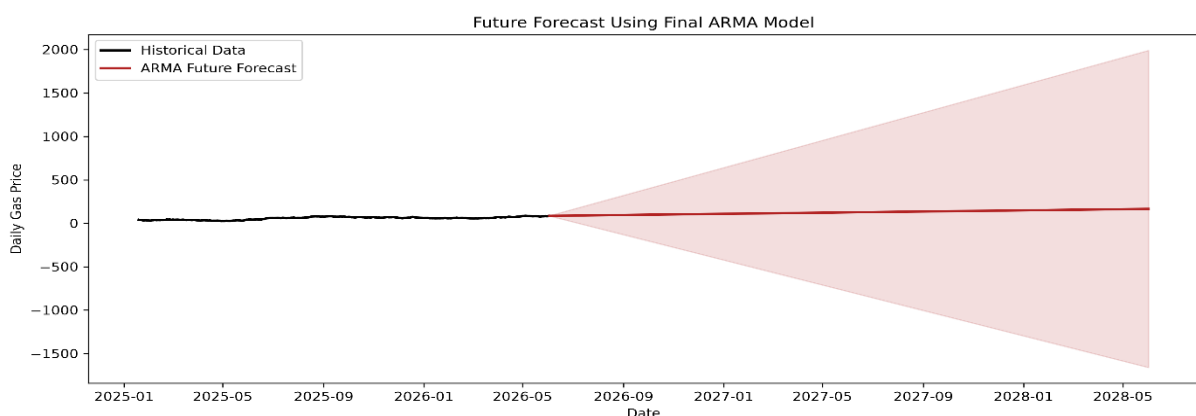


Figure 9. Twenty-Four Month Forecast Generated Using Final ARMA Model

The forecast indicates that gas prices are likely to remain relatively stable over the next 24 months, although a slight upward trend can be observed. As the forecast moves further into the future, the prediction intervals become wider, suggesting greater uncertainty in the projected values. Therefore,

while the forecast provides an indication of possible future price movements, actual gas prices may differ from these estimates.

It is important to note that the forecasts were generated solely from historical gas price data. Consequently, they should be interpreted as indicative trends rather than exact predictions, particularly because future prices may be influenced by external factors that were not included in the analysis.

6.3 Limitations

A few limitations should be considered when interpreting these findings. The analysis was based on approximately sixteen months of data, which may not fully capture longer-term changes in gas price behaviour. In addition, both models relied only on historical price information and did not consider external influences such as weather conditions, economic developments, or geopolitical events that can affect energy markets. While ARMA produced the most accurate forecasts for this dataset, its performance may differ when applied to data with stronger trends or seasonal patterns.

Future studies could improve forecasting accuracy by incorporating external variables and evaluating other approaches, such as SARIMA, machine learning, or hybrid forecasting models.

7. Conclusion

During the initial data exploration, it became evident that the gas price series was not stationary. Therefore, first-order differencing was used before developing the forecasting models.

Using an AIC-based grid search, the optimal specification was identified as (5,1,8), and the final forecasting model was implemented as ARMA(5,8) on the differenced series. When evaluated on unseen data, ARMA achieved lower forecasting errors than Prophet across all evaluation metrics, demonstrating stronger predictive performance for this dataset.

The findings suggest that short-term patterns in the gas price series played a greater role in forecasting accuracy than trend-based modelling approaches. The future forecasts suggested relatively stable gas price behaviour over the forecast horizon, although the level of uncertainty increased substantially as the prediction period extended further into the future.

Overall, the ARMA(5,8) model provided the most accurate forecasts and was therefore selected as the preferred forecasting approach. Future studies could improve forecasting accuracy by incorporating external variables and exploring more advanced forecasting techniques.

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